UNIT-1

Q1. E-commerce

* E-commerce is electronic commerce
* E-commerce combines the advantages of computer based processing (analyzing, reviewing, etc) and human based processing (Ideas, thinking, etc).
* E-commerce is used for analyzing, thinking, reviewing of the information.
* These have 3 tiers:

→ Tier 1: Mainly for Advertising

→ Tier 2: It also contains tier-1 and used for giving information among the Users.

→ Tier 3: It contains both Tier 1 and Tier 2 and is also used most frequently.

* E-commerce have few applications. They are:

(1) Electronic fund Transfer
(2) Enterprise integration
(3) computer support collaborative work
(4) Government regulatory data interchange.
(5) Books
(6) Travel services
(7) Automobile specification (purchase, selling, time table)
(8) Flowers

(9) computers

(10) EDI services

(11) Advertising

(12) Magzines

(13) Banking

(14) Online shopping malls.

*Available communication apparatus in e-commerce environment:

⇒ The communication between the buyer and seller will be there.

⇒ The buyers can directly communicate with the sellers if there are any queries or faults.

⇒ cost will be reduced for buyers as there will be lot of options for a product.

⇒ cost will be reduced also for the sellers as there will be many competitors.

⇒ Inventories will also be reduced as there is no concept of wastage in E-commerce.

⇒ There will be many new opportunities for people.

⇒ In e-commerce, there will be high speed of analyzing, reviewing, etc.

⇒ Many people can start their own bussiness through E-commerce.

→ The quality and quantity of the product will also be up-to the mark.

→ As, buyers can directly speak or communicate with the merchants, there doesn't exist any queries remained unasked for the buyers.

→ Starting their own business have become easier because of E-commerce.

→ Everything will be clean and clear for both buyers and sellers.

→ The products or items will be directly delivered to the buyer or customer.

→ Online transactions has also become much easier because of E-commerce.

→ In most cases, transactions are being successfully completed.

* This is E-commerce and these are some of the available communication apparatus and some of the benefits in E-commerce.

UNIT-II

Q3 SHTTP and SSL protocols

SHTTP

* SHTTP means Secure hyper text transfer protocol
* SHTTP gives Security to the information and online transactions.
* SHTTP is a secure extension of HTTP (hyper text transfer protocol)
* This follows RSA protocols for securing the information.
* This adds public key in addition to the normal secure password.
* There are two types of keys in this protocol. They are
  i) public key
  ii) private key.
* A company produces these two keys.
* When a company want to send a file to other company, it uses the public key and retain the private key.
* The file will be sent to the company and the receipient receives it.
* This receipient uses private key to access the information (or) file.
* This is how SHTTP works by using two

Keys.

* These measures are for securing the information without any unauthorized access.
* SHTTP also ensures the Online transactions without any faults.

SSL Protocol

* SSL protocol means Secure Socket layer protocol.
* SSL lies between the Tcp/Ip and Application layer.
* It follows RSA protocols for securing the Online transactions.
* Through SSL protocol, we get:
  - ⇒ Authentication and non-repudiation for servers, by using digital signatures.
  - ⇒ Authentication and non-repudiation for customers by using digital signatures.
  - ⇒ It ensures Confidentiality
  - ⇒ It ensures Authentication and integrity.
* The Online transactions will be done through many security measures.
* The card number given by the card holder will be sent to the merchant.
* The merchant will analyze it and further

continues the transaction process.

* The SSL protocol works through,

=> SSL protocol encrypts the information for the security of the data.

=> SSL protocol Authorises the users to check if they are authorized users are not. It checks both the users.

=> SSL protocols ensure integrity, as there is no information altered which may take place when information is exchanged.

* This is the working of SSL protocol.

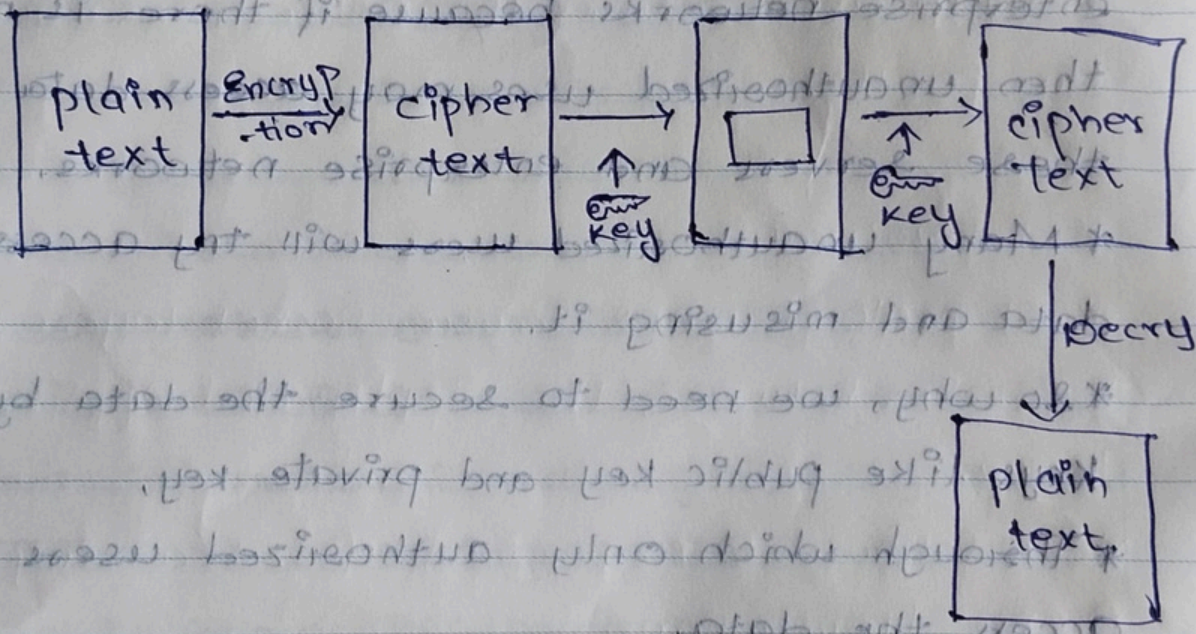
* As it follows RSA protocol, it ensures utmost security for information on the Online transaction.

UNIT-III

Q5. Need of Security on Web Services and enterprise networks

* Security is needed for the web servers and enterprise networks because if there is no security then unauthorized uses may access data from these servers and enterprise networks.
* Many unauthorized users will try accessing the data and misusing it.
* So why, we need to secure the data by adapting keys like public key and private key,
* Through which only authorized users can access the data.
* For securing the data, first we need to encrypt the data as other users cant understand it.
* This encrypted data must be secured by using a key.
* The receiver will have another key which only both the parties know.
* When the receiver want to access the data, he will use the private key and access it.
* He then converts the encrypted information into normal plain text by performing decryption on the encrypted data.

* Through this the receiver can accen the data or information without any unauthorized accen



* This is why security is needed on web servers and enterprise networks.
* Enterprise networks needs security as the data from the enterprise may be leaked.